

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\gamma_{g,9} : |D_{\text{inv}}\rangle \rightarrow \frac{1}{\sqrt{D}} \sum_{\alpha} \sum_{\beta, \gamma} c_{\alpha\beta} \bar{c}_{\gamma\alpha} |d_{\beta} \otimes \bar{d}_{\gamma}\rangle = \frac{1}{\sqrt{D}} \sum_{\alpha} |d_{\alpha} \otimes \bar{d}_{\alpha}\rangle = |D_{\text{inv}}\rangle ,$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$X(\alpha_1, \alpha_2, \alpha'_1, \alpha'_2) = \exp \left[i\alpha_1 q^1 \frac{\partial}{\partial q^1} + i\alpha_2 q^2 \frac{\partial}{\partial q^2} + i\alpha'_1 \bar{q}^1 \frac{\partial}{\partial \bar{q}^1} + i\alpha'_2 \bar{q}^2 \frac{\partial}{\partial \bar{q}^2} \right]$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\Gamma_{ij}^{\mathbf{t}} = a \left(\frac{da}{d\mathbf{t}} \right) \gamma_{ij} , \quad \Gamma_{\mathbf{t}j}^i = \frac{1}{a} \left(\frac{da}{d\mathbf{t}} \right) \delta_j^i , \quad \Gamma_{jk}^i = \frac{\gamma^{il}}{2} (\partial_j \gamma_{lk} + \partial_k \gamma_{lj} - \partial_l \gamma_{jk}) .$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$lA_1 + nB_1 = lU_l^n Str \left(\dots \phi^{i'_{2k-1}} \dots \underbrace{\partial_{a_l} \phi^{j_l} \phi^{i'_{2k}}}_{\dots} \dots \right) + nU_l^n Str \left(\dots \phi^{i'_{2k-1}} \dots \underbrace{\phi^{i_n} \phi^{i'_{2k}}}_{\dots} \dots \partial_{a_l} \phi^{j_l} \dots \right)$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\Lambda_{\pm}(q_0) = \Lambda(q_0) \pm \frac{2\kappa \left(\frac{2}{\pi} \right)^{\frac{1}{2}}}{[\frac{1}{2}(q_0 - 1)]!} \left(\frac{1+k}{1-k} \right)^{-\frac{\kappa}{k}} \left(\frac{8\kappa}{1-k^2} \right)^{\frac{1}{2}q_0} \left[1 + O\left(\frac{1}{\kappa} \right) \right]$$